

2017-Ph-H1-Q14

Section: Our Dynamic Universe

Topic: Equations and Graphs

An object is dropped. Time and distance are given. Find its speed after 2 s.

Use:

$$v = u + at$$

Here:

- $u = 0$
- $a = 9.8$
- $t = 2.0$

$$v = 0 + 9.8 \times 2 = 19.6 \text{ m/s}$$



Final Answer: A — 19.6 m/s



Revision Tips:

- For vertical motion:

$$v = u + at, \quad s = ut + \frac{1}{2}at^2$$

- Objects fall with acceleration 9.8 m/s^2